

Hari Narayan Sah

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PROFESSIONAL SUMMARY

AI Engineer specializing in Voice AI and Computer Vision, with hands-on experience building production voice agents, real-time interview and scribe systems, and localized Nepali speech models. Experienced in developing scalable WebSocket pipelines, AI automation workflows, speech-processing systems, and high-accuracy OCR solutions.

SKILLS

- **Voice AI:** Amazon Nova Sonic, Amazon Transcribe, Vapi, LiveKit, Twilio, n8n, Silero VAD, RNNoise, Azure AI Speech, ElevenLabs, GroqSTT, Nepali TTS fine-tuning (Chatterbox, Matcha-TTS)
- **LLM / Agents:** AWS Strands, Claude Haiku/Sonnet on Amazon Bedrock, LangChain, Groq, RAG, FAISS, tool calling
- **ML/DL & Vision:** PyTorch, TensorFlow, Transformers, XGBoost, Roboflow, OpenCV, ONNX Runtime, CLIP
- **Cloud:** AWS (Amazon Bedrock, Amazon Bedrock AgentCore, S3, DynamoDB, ECR, Lambda), Azure, Docker
- **Backend:** FastAPI, WebSockets, JavaScript
- **Data:** Pinecone, Chroma, SQL
- **Languages:** Python, C, C++, JavaScript

EXPERIENCE

AI Engineer | Mantra Ideas, Lalitpur | Sep 2025 – Present

Colait: clinical voice interview and scribe platform

- Built the production voice interview runtime: FastAPI WebSocket service with Nova Sonic for speech and Claude Haiku 4.5 on Amazon Bedrock (Strands) for probing, transitions, and structured answers (choice, Likert, ranking)
- Made sessions resilient on poor networks with quiet-moment reconnects, replay of unspoken speech, heartbeat resume, and real-time transcript writes to S3
- Shipped chat-only interviews and mid-session voice-to-chat switching on the same question state
- Built Colait Scribe: streaming Amazon Transcribe with speaker diarization, Haiku-generated visit briefs, and romanized transcripts
- Built the anonymized interview corpus behind analytics and PDF reports, with PII scrubbing before report generation
- Deployed services as ARM64 images to ECR and Amazon Bedrock AgentCore

AnswerServiceAI: client voice agents (UK)

- Built voice agents for hotel order management and garage bookings using Vapi, n8n, and Twilio, including live workflow automation and handoff

Data Science Intern | Mindrisers Technologies, Kathmandu | June 2025 – Sep 2025

- Developed generative AI assistants with LLaMA and LangChain, including a medical chatbot and a news research app (Groq Llama 3 and FAISS)
- Trained churn prediction and financial sentiment models and deployed them with the engineering team

PROJECT

TelvoxAI

- Built a voice AI platform from scratch, similar to Vapi, on LiveKit with multi-tool agents (leads, calendar, transfer, custom HTTP tools) and Azure AI Speech, ElevenLabs, Google, OpenAI, and Groq STT integrations
- Implemented background noise reduction (Silero VAD, RNNoise, WebRTC noise gain) and cut latency from 8–9 sec to 1–2 sec by moving STT to Groq
- Fine-tuned Chatterbox for Nepali (Devanagari), built Matcha-TTS Nepali inference, and created multiple Nepali voice agents

BanknoteOCR

- Built a pipeline that identifies Nepali banknote denominations and signatures using a Roboflow-trained detector and a PyTorch siamese signature model, served via FastAPI
- Achieved approximately 90% accuracy in denomination and signature identification

EDUCATION

- Data Science with Python Certification – Mindrisers Institute of Technology,
- Kathmandu (Sep 2024 – Jan 2025): [Credential Link](#)
- Diploma in Computer Engineering – Adarsha Secondary School, Biratnagar (2020 – 2024)
- Diploma in Graphic Designing – New BIIT Institute, Biratnagar (Jan 2023 – Apr 2023)

LANGUAGES

Maithili (Native) | Nepali (Fluent) | Hindi (Fluent) | English (Professional Working Proficiency)